Name: ______________________ Homework 8 | Math 253 | Huchala

Due Wednesday of Week 10 at the start of class

Complete the following problems and submit them in class. 8 points are awarded for thoroughly attempting every problem, and I'll select four problems to grade on correctness. Enough work should be shown that there is no question about the mathematical process used to obtain your answers.

In problems 1–5, find a Taylor series for $f(x)$ centered at $x = a$ and determine its interval of convergence.

1. $f(x) = \cos(x)$, $a = 0$.

2. $f(x) = \sin(2x)$, $a = \frac{\pi}{2}$.

3. $f(x) = \ln(-x)$, $a = -1$.

4. $f(x) = \frac{1}{x}$, $a = 1$.

5. $f(x) = \sqrt[3]{x+1}$, $a = 0$.

In problems 6–9, compute the value of the series and justify your answer.

6. $\displaystyle\sum_{n=0}^{\infty} \frac{2^n}{n!}$.

7. $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{n+1}$.

8. $\displaystyle\sum_{n=0}^{\infty} (n+1)\left(\frac{1}{2}\right)^n$.

9. $\displaystyle\sum_{n=0}^{\infty} \frac{(-4)^n}{(2n)!}$.

In problems 10–13, approximate the value to within 0.01 using Taylor series.

10. $\cos(2)$.

11. $\frac{1}{e^2}$.

12. $\ln\left(\frac{1}{2}\right)$.

13. $\sqrt[4]{\frac{1}{2}}$.

14. Using the fact that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, evaluate $\int_0^1 \frac{\ln(1+x)}{x}$.

15. Find a series solution to $\frac{1}{x} + y'' = \frac{y}{x}$.

16. The function $\sin(x)$ has zeroes at exactly $x = \pi n$ for integers n, and this allows us to write $\sin(x)$ as

$$\sin(x) = x\left(1 - \frac{x}{\pi}\right)\left(1 + \frac{x}{\pi}\right)\left(1 - \frac{x}{2\pi}\right)\left(1 + \frac{x}{2\pi}\right)\cdots.$$

Note that this isn't true for functions in general! There can often be a lot more going on with a function than we can determine by its zeros.

a) Divide by x to find an expression for $\dfrac{\sin(x)}{x}$.

b) Use the difference of squares formula $(a - b)(a + b) = a^2 - b^2$ to group each two consecutive factors in the product.

c) Expand $\dfrac{\sin(x)}{x}$ as a Maclaurin series. What is the coefficient of x^2?

d) In the grouped product, what is the coefficient of x^2? You'll likely need to express it as a sum.

e) Set the two sides equal to one another. What have you shown?